Dmitry Ivanov

Senior Frontend Engineer

Frontend engineer focused on interactive graphics and high-performance web applications. Built scientific visualization at Plotly, fintech products at Mobeewave, Apple, and Brainfinance, and voice interfaces at ElevenLabs. Current open-source work covers interactive tools, WebGL, color, and audio. Trained in control systems and signal processing.

CONTACT

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WORK

Software Engineer — Oct 2025 – Present

Independent

Open-source graphics, audio, and WebAssembly; selected projects total 3,000+ GitHub stars.

- color-space.io: interactive atlas of 205 color spaces, each tested against a cited reference
- settings-panel: UI controls inferred from plain values
- gl-waveform: WebGL2 waveform renderer; 60 fps pan and zoom over 172.8M samples
- audio: audio editing in JavaScript; 300+ GitHub stars
- JZ / watr: JavaScript-to-WebAssembly compiler
- Claude Code daily; agent changes verified by differential tests against reference implementations

Senior Frontend Engineer — Nov 2022 – Oct 2025

Brainfinance

Consumer fintech serving 100,000+ users in Canada: icash.ca loans and earna.zone savings.

- Built the earna.zone design system and frontend on Next.js (shadcn, Tailwind, Storybook): landing, dashboard, application flows
- Rebuilt icash.ca loan application and contract signing; fixed validation errors that blocked form submission
- Raised key flows above 95 in PageSpeed; fixed WCAG AA accessibility issues

Frontend Engineer (Contract) — Oct 2022 – Nov 2022

ElevenLabs

Pre-seed voice AI contract.

- Launched elevenlabs.io: 11ty, Firebase auth, Stripe payments, voice API access
- Built the browser player for generated speech

Software Engineer — Jan 2019 – Oct 2022

Mobeewave / Apple

Joined Mobeewave in 2019 and continued at Apple after the 2020 acquisition.

- Mobeewave: Samsung POS tap-to-pay frontend (PCI DSS), shipped in Samsung Wallet; acquirer portal used by 50+ financial institutions
- Apple: Wallet web push provisioning, Apple Business Registry portal, Pay Later checkout widget
- Reference polyfill for the W3C Template Instantiation proposal

WebGL Engineer — Sep 2016 – Sep 2018

Plotly

Built and maintained the WebGL stack for plotly.js, an open-source scientific charting library.

- WebGL scatter, line, scatterplot-matrix, and waveform renderers
- Point clustering for real-time exploration of up to 1 billion points
- Visual-regression testing across 2,000+ chart configurations
- Maintained regl (5,500+ GitHub stars) and 24 gl-vis repositories

OPEN SOURCE

Wavefont / Linefont

Variable fonts that draw waveforms and line charts as text; on Google Fonts; 900+ GitHub stars combined.

AudioJS

70+ audio and DSP repositories for browsers and Node.

- web-audio-api: Web Audio API for Node; 950+ GitHub stars
- digital-filter: IIR, FIR, adaptive, and multirate filters

sprae / Spect

sprae: reactive HTML attributes; made earna.zone server components interactive without client components. Spect: state and component framework.

EARLIER EXPERIENCE

Web Developer — Jan 2015 – Aug 2016

TTBA Group

10+ sites at 100/100 PageSpeed; Wishbox.gift on Preact.

Frontend Developer — Mar 2012 – Jan 2015

KudaGo

City guide for 15+ cities; mobile site under 20 KB of JavaScript, 100/100 PageSpeed at launch.

EDUCATION

Baltic State Technical University "Voenmeh" — Sep 2006 – Jun 2012

BSc + MSc

BSc: Information & Computer Science; MSc: Computer Graphics & Intelligent Systems

Saint Petersburg College of Marine Instrumentation — Sep 2002 – Jun 2006

Technician diploma

Automatic Control Systems · Signal Processing · Control Theory · Electronics · Instrumentation

SKILLS

Interactive graphics

Canvas · SVG · WebGL · regl · GLSL · variable fonts · pointer & touch input · requestAnimationFrame

Frontend

TypeScript · JavaScript · React · Next.js · Preact · signals · web components · SSR

Performance

profiling · Web Workers · OffscreenCanvas · WebAssembly · Core Web Vitals · Lighthouse

Testing

differential tests · visual regression · Playwright · Puppeteer · Jest

Audio & DSP

Web Audio API · FFT · IIR/FIR filters · codecs · streaming

Tools & platform

Claude Code · MCP · Figma plugins · Node.js · Go · Python · GitHub Actions · esbuild / Rollup